

Section 1 - Identification

Product Name

HI Broth with BCP and with/without supplement

Product Code

R061052, R061056, R061060, R061064, R061072, R061080, R061084, R061088, R061092, R061096, R061100, R061104, R061108, R061112, R061116, R061120, R061121, R061128, R061129, R061132, R061136, R061140

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Recommended Use

Laboratory chemicals.

Uses advised against

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

Section 2 - Hazard(s) Identification

Classification under Safe Work Australia

Classified as not hazardous according to criteria of Safe Work Australia.

Physical hazards

No hazards identified

Health hazards

No hazards identified

Environmental hazards

No hazards identified

Label Elements

None required

Other information

This product does not contain any known or suspected endocrine disruptors

Section 3 - Composition and Information on Ingredients

Component	CAS No	Weight %
Water	7732-18-5	97.41
Phenol, 4,4'-(3H-2,1-benzoxathiol-3-ylidene)bis[2-bromo-6-methyl- S,S-dioxide, monosodium salt	62625-30-3	Trace
Inulin	9005-80-5	0-1
L-Arabinopyranose	87-72-9	0-1
Glycerin	56-81-5	0-1
D-Galactose	59-23-4	0-1
Galactitol	608-66-2	0-1
D-Glucose, 4-O-.beta.-D-glucopyranosyl-	528-50-7	0-1
Ribitol	488-81-3	0-1
myo-Inositol	87-89-8	0-1
Trehalose dihydrate	6138-23-4	0-1
D-Xylose	58-86-6	0-1
Sucrose	57-50-1	0-1
L-Sorbose	87-79-6	0-1
D-Glucitol	50-70-4	0-1
D-Ribose	50-69-1	0-1
L(+)-Rhamnose monohydrate	10030-85-0	0-1
Raffinose pentahydrate	17629-30-0	0-1
D-Glucose, 6-O-.alpha.-D-galactopyranosyl-	585-99-9	0-1
D-Mannose	3458-28-4	0-1
D(+)-Maltose monohydrate	6363-53-7	0-1
.beta.-D-Lactose	63-42-3	0-1
D-Mannitol	69-65-8	0-1

Section 4 - First Aid Measures

Inhalation	Remove to fresh air.
Ingestion	Clean mouth with water and drink afterwards plenty of water.
Skin Contact	Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.
Eye Contact	Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a physician.
Self-Protection of the First Aider	No special precautions required.
First Aid Facilities	Eyewash, safety shower and washroom.
Most important symptoms and effects	No information available.
Notes to Physician	Treat symptomatically.

Section 5 - Fire Fighting Measures

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Section 6 - Accidental Release Measures

Emergency procedures

Ensure adequate ventilation.

Environmental Precautions

See Section 12 for additional Ecological Information.

Methods for Containment and Clean Up

Clean-up methods - small spillage

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal.

Clean-up methods - large spillage

Typically only supplied is small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

Section 7 - Handling and Storage

Precautions for Safe Handling

Ensure adequate ventilation.

Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

Section 8 - Exposure Controls and Personal Protection

Exposure limits

AUS - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

ACGIH - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

UK - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

DE - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

NZ - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

Component	Australia	New Zealand WEL	ACGIH TLV	The United Kingdom	Germany
Glycerin	TWA: 10 mg/m ³	TWA: 10 mg/m ³		TWA: 10 mg/m ³ 8 hr (mist only)	TWA: 200 mg/m ³ (8 Stunden). AGW - exposure factor 2

					TWA: 200 mg/m ³ (8 Stunden). MAK Höhepunkt: 400 mg/m ³
Sucrose	TWA: 10 mg/m ³	TWA: 10 mg/m ³	TWA: 10 mg/m ³	STEL: 20 mg/m ³ 15 min TWA: 10 mg/m ³ 8 hr	

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Exposure Controls

Engineering Measures

None under normal use conditions.

Personal protective equipment

Eye Protection

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	AUS/NZ Standard	Glove comments
Disposable gloves	See manufacturers recommendations	-	AS/NZS 2161	(minimum requirement)

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Skin and body protection

Long sleeved clothing

Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

Recommended Filter type:

Particle filter (or AUS/NZ equivalent)

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

Environmental exposure controls

No information available.

Section 9 - Physical and Chemical Properties

Information on basic physical and chemical properties

Appearance

Physical State

Liquid

Odor

No information available

Odor Threshold

No data available

pH

No information available

Melting Point/Range

No data available

Softening Point

No data available

Boiling Point/Range

No information available

Flash Point

No information available

Method - No information available

Evaporation Rate	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Vapor Pressure	No data available	
Vapor Density	No data available	(Air = 1.0)
Specific Gravity / Density	No data available	
Bulk Density	Not applicable	Liquid
Water Solubility	No information available	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Glycerin	-1.75	
myo-Inositol	-2.08	
Sucrose	-3.67	
D-Glucitol	-2.2	
D-Ribose	-2.32	
D-Mannose	-2.71	
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	No data available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Other information

VOC Content(%) 1

Section 10 - Stability and Reactivity

Reactivity	None known, based on information available
Stability	Stable under normal conditions.
Conditions to Avoid	Heat, flames and sparks.
Incompatible Materials	None known.
Hazardous Decomposition Products	None under normal use conditions.
Hazardous Polymerization	No information available.

Section 11 - Toxicological Information

Information on Toxicological Effects

Product Information

(a) acute toxicity;

Oral	Based on available data, the classification criteria are not met
Dermal	Based on available data, the classification criteria are not met
Inhalation	Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	-	-	-
Glycerin	12600 mg/kg (Rat)	> 10 g/kg (Rabbit)	> 2.75 mg/L/4h (Rat)(mist)
Trehalose dihydrate	4600 mg/kg (Rat)		

Sucrose	LD50 = 29700 mg/kg (Rat)		
D-Glucitol	LD50 = 15900 mg/kg (Rat)		
D-Mannose	> 5000 mg/kg (Rat)	> 5000 mg/kg (Rat)	
.beta.-D-Lactose	LD50 > 10 g/kg (Rat)		
D-Mannitol	LD50 = 13500 mg/kg (Rat)		

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects,both acute and delayed No information available

Section 12 - Ecological Information

Ecotoxicity effects

Component	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
L-Arabinopyranose	LC50: = 37700 mg/L, 96h static (Pimephales promelas)			
Glycerin	LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss)			

Persistence and Degradability No information available

Bioaccumulative Potential No information available

Component	log Pow	Bioconcentration factor (BCF)
Glycerin	-1.75	No data available
myo-Inositol	-2.08	No data available
Sucrose	-3.67	No data available
D-Glucitol	-2.2	No data available
D-Ribose	-2.32	No data available
D-Mannose	-2.71	No data available

Mobility	No information available.
Endocrine Disruptor Information	This product does not contain any known or suspected endocrine disruptors
Persistent Organic Pollutant	This product does not contain any known or suspected substance
Ozone Depletion Potential	This product does not contain any known or suspected substance

Section 13 - Disposal Considerations

Waste from Residues/Unused Products	Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.
Contaminated Packaging	Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.
Other Information	Chemical wastes should be disposed through a licensed commercial waste collection service.

Section 14 - Transport Information

IMDG/IMO	Not regulated
ADG	Not regulated
IATA	Not regulated
Environmental hazards	No hazards identified
Special Precautions	No special precautions required
Additional information	None known

Section 15 - Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

National Regulations **Australia**

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons

No poison schedule number allocated.

Australian Industrial Chemicals Introduction Scheme (AICIS)

Component	Australian Industrial Chemicals Introduction Scheme (AICIS)	Additional information
Water - 7732-18-5	Present	-
Inulin - 9005-80-5	Present	-
L-Arabinopyranose - 87-72-9	Present	-

Glycerin - 56-81-5	Present	-
D-Galactose - 59-23-4	Present	-
Galactitol - 608-66-2	Present	-
D-Glucose, 4-O-.beta.-D-glucopyranosyl- - 528-50-7	Present	-
Ribitol - 488-81-3	Present	-
myo-Inositol - 87-89-8	Present	-
Trehalose dihydrate - 6138-23-4	Present	-
D-Xylose - 58-86-6	Present	-
Sucrose - 57-50-1	Present	-
L-Sorbose - 87-79-6	Present	-
D-Glucitol - 50-70-4	Present	-
D-Ribose - 50-69-1	Present	-
L(+)-Rhamnose monohydrate - 10030-85-0	Present	-
Raffinose pentahydrate - 17629-30-0	Present	-
D-Glucose, 6-O-.alpha.-D-galactopyranosyl- - 585-99-9	Present	-
D-Mannose - 3458-28-4	Present	-
D(+)-Maltose monohydrate - 6363-53-7	Present	-
.beta.-D-Lactose - 63-42-3	Present	-
D-Mannitol - 69-65-8	Present	-

Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Not applicable

Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

International Inventories

Component	AICS	NZIoC	EINECS	ELINCS	TSCA	DSL	NDSL	PICCS	ENCS	ISHL	IECSC	KECL
Water	X	X	231-791-2	-	X	X	-	X	X		X	KE-35400
Phenol, 4,4'-(3H-2,1-benzoxathiol-3-ylidene)bis[2-bromo-6-methyl-, S,S-dioxide, monosodium salt	-	-	263-655-3	-	X	X	-	-	-		X	-
Inulin	X	X	232-684-3	-	X	X	-	X	-	X	X	-
L-Arabinopyranose	X	X	201-767-6	-	X	-	X	X	-		-	KE-01896
Glycerin	X	X	200-289-5	-	X	X	-	X	X	X	X	KE-29297
D-Galactose	X	X	200-416-4	-	X	X	-	X	X	X	X	KE-17418
Galactitol	X	X	210-165-2	-	X	X	-	-	-		X	-
D-Glucose, 4-O-.beta.-D-glucopyranosyl-	X	X	208-436-5	-	X	X	-	X	-	X	X	-
Ribitol	X	X	207-685-7	-	X	X	-	X	-	X	-	-
myo-Inositol	X	X	201-781-2	-	X	X	-	X	X	X	X	KE-21013
Trehalose dihydrate	X	X	-	-	-	-	-	X	-	X	X	-
D-Xylose	X	X	200-400-7	-	X	X	-	X	X	X	X	KE-35439
Sucrose	X	X	200-334-9	-	X	X	-	X	-	X	X	KE-17258
L-Sorbose	X	X	201-771-8	-	X	X	-	X	X	X	X	KE-31710

D-Glucitol	X	X	200-061-5	-	X	X	-	X	X	X	X	KE-31708
D-Ribose	X	X	200-059-4	-	X	X	-	X	X	X	X	KE-30342
L(+)-Rhamnose monohydrate	X	X	-	-	-	-	-	-	-	X	X	-
Raffinose pentahydrate	X	X	-	-	-	-	-	-	-		X	-
D-Glucose, 6-O-.alpha.-D-galactopyranosyl-	X	X	209-568-6	-	X	X	-	-	-		X	-
D-Mannose	X	X	222-392-4	-	X	X	-	X	X	X	X	KE-23062
D(+)-Maltose monohydrate	X	X	-	-	-	-	-	X	-		X	-
.beta.-D-Lactose	X	X	200-559-2	-	X	X	-	X	-	X	X	KE-17416
D-Mannitol	X	X	200-711-8	-	X	X	-	X	X	X	X	KE-23061

Legend: X - Listed. '-' - Not Listed. XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)). **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

International Regulations

Ozone Depletion Potential This product does not contain any known or suspected substance

Persistent Organic Pollutant This product does not contain any known or suspected substance

Rotterdam Convention (PIC) Not applicable

Basel convention on the control of transboundary movements of hazardous wastes and their disposal
Not applicable.

Component	CAS No	OECD HPV	Restriction of Hazardous Substances (RoHS)	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Water	7732-18-5	Listed	Not applicable	Not applicable	Not applicable
Phenol, 4,4'-(3H-2,1-benzoxathiol-3-ylidene)bis[2-bromo-6-methyl-, S,S-dioxide, monosodium salt	62625-30-3	Not applicable	Not applicable	Not applicable	Not applicable
Inulin	9005-80-5	Not applicable	Not applicable	Not applicable	Not applicable
L-Arabinopyranose	87-72-9	Not applicable	Not applicable	Not applicable	Not applicable
Glycerin	56-81-5	Listed	Not applicable	Not applicable	Not applicable
D-Galactose	59-23-4	Not applicable	Not applicable	Not applicable	Not applicable
Galactitol	608-66-2	Not applicable	Not applicable	Not applicable	Not applicable
D-Glucose, 4-O-.beta.-D-glucopyranosyl-	528-50-7	Not applicable	Not applicable	Not applicable	Not applicable
Ribitol	488-81-3	Not applicable	Not applicable	Not applicable	Not applicable
myo-Inositol	87-89-8	Not applicable	Not applicable	Not applicable	Not applicable
Trehalose dihydrate	6138-23-4	Not applicable	Not applicable	Not applicable	Not applicable
D-Xylose	58-86-6	Not applicable	Not applicable	Not applicable	Not applicable
Sucrose	57-50-1	Listed	Not applicable	Not applicable	Not applicable
L-Sorbose	87-79-6	Not applicable	Not applicable	Not applicable	Not applicable
D-Glucitol	50-70-4	Listed	Not applicable	Not applicable	Not applicable
D-Ribose	50-69-1	Not applicable	Not applicable	Not applicable	Not applicable
L(+)-Rhamnose monohydrate	10030-85-0	Not applicable	Not applicable	Not applicable	Not applicable
Raffinose pentahydrate	17629-30-0	Not applicable	Not applicable	Not applicable	Not applicable
D-Glucose, 6-O-.alpha.-D-galactopyranosyl-	585-99-9	Not applicable	Not applicable	Not applicable	Not applicable

D-Mannose	3458-28-4	Not applicable	Not applicable	Not applicable	Not applicable
D(+)-Maltose monohydrate	6363-53-7	Not applicable	Not applicable	Not applicable	Not applicable
.beta.-D-Lactose	63-42-3	Not applicable	Not applicable	Not applicable	Not applicable
D-Mannitol	69-65-8	Not applicable	Not applicable	Not applicable	Not applicable

Authorisation/Restrictions according to EU REACH

Not applicable

Section 16 - Other Information

Legend

AICS - Australian Inventory of Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List
IECSC - Chinese Inventory of Existing Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
NZS 5433:2020 - Transport of Dangerous Goods on Land
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
WEL - Workplace Exposure Limit
DNEL - Derived No Effect Level
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative
VOC - (Volatile Organic Compound)

NZIoC - New Zealand Inventory of Chemicals
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
ENCS - Japanese Existing and New Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
CAS - Chemical Abstracts Service
ACGIH - American Conference of Governmental Industrial Hygienists
PNEC - Predicted No Effect Concentration
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
ADG - Australian Code for the Transport of Dangerous Goods by Road and Rail
OECD - Organisation for Economic Co-operation and Development
LC50 - Lethal Concentration 50%
ATE - Acute Toxicity Estimate
RPE - Respiratory Protective Equipment
NOEC - No Observed Effect Concentration
BCF - Bioconcentration factor
PBT - Persistent, Bioaccumulative, Toxic

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data
Health Hazards Calculation method
Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Revision Date 05-Jul-2023
Revision Summary Not applicable.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet